

Group II, Row IV — checked and finished

Where $d_\perp$ goes, what is wrong in (0.1)–(0.3), and the contraction and p^+ integration carried out

$$S = p^+ + k^+, \quad \sigma = \frac{S}{k^+}, \quad r = y' - y, \quad \kappa = k_\perp \cdot r, \quad X^k = \frac{(x - y)^k}{(x - y)^2}, \quad X'^{k'} = \frac{(x' - y')^{k'}}{(x' - y')^2}, \quad \mathcal{L} = \frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu_{\text{MS}}^2 r^2)$$

Short answer on $d_\perp$. It cancels *exactly* inside $\mathcal{G}^{mm'}$ — the $\frac{1}{d_\perp}$ of your (0.12) is eaten by the $\frac{d}{2}$ of master (0.13), so $\frac{1}{d_\perp}(\frac{1}{\epsilon} - 1) = \frac{1}{2\epsilon}$ with no remainder. It then **comes back twice at the contraction** in (0.21), and there it does *not* cancel: the explicit $d_\perp$ in the first bracket term turns $\mathcal{L} \rightarrow \mathcal{L} - 1$, and the trace $\delta^{mm'} \delta^{mm'} = d_\perp$ in the third term turns $\mathcal{L} \rightarrow \mathcal{L} - 2$. Setting $d_\perp = 2$ too early loses both.

1. Status of every step

Step	Status	Comment
(0.1) tensor product, summed over i, j	correct	gives $[d_\perp - 2a - 2b]\delta^{km}\delta^{k'm'} + [a^2 + b^2]\delta^{mm'}\delta^{kk'} + 2ab\delta^{km'}\delta^{mk'}$ with $a = \frac{S}{p^+}, b = \frac{S}{k^+}$; checked in $D = 3, 4, 5$ to machine zero
(0.2) <i>relative</i> weights in the bracket	correct	all three match exactly
(0.2) overall $\frac{(k^+ + p^+)}{k^+}$	wrong	spurious — the prefactor should be just $\frac{1}{k^+}$ (see §2)
(0.2) the phase	wrong	$e^{-ik \cdot (y' - y)} \rightarrow e^{-i\frac{S}{k^+} k \cdot (y' - y)}$ — and this matters a lot (§6)
(0.4)–(0.11) Feynman parameter, shift, odd terms	correct	$u^m = [l - (1 - x)r]^m, (u + r)^{m'} = [l + xr]^{m'} \checkmark, \Delta = x(1 - x)r^2 \checkmark$
(0.12)–(0.17) masters, x integral	correct	$\int_0^1 [x(1 - x)]^{-\epsilon} dx = B(1 - \epsilon, 1 - \epsilon) = \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)} \checkmark$
(0.18) use of $\Gamma[\epsilon] = \Gamma[1 + \epsilon]/\epsilon$	correct	$(1 - \epsilon)\Gamma[\epsilon] = \Gamma[1 + \epsilon](\frac{1}{\epsilon} - 1) \checkmark$
p.3 gamma expansions	correct	$\Gamma[1 + \epsilon] = 1 - \gamma_E \epsilon \checkmark$ and $\frac{\Gamma[1 - \epsilon]^2}{\Gamma[2 - 2\epsilon]} = 1 + 2\epsilon \checkmark$ (product = $1 + (2 - \gamma_E)\epsilon$)
(0.19) $\mu^{2\epsilon}$ vs (0.20) log	inconsistent	these two cannot both hold; see §4
(0.3) bookkeeping	typos	phase written twice; $\frac{p^+}{k^+}$ appears where (0.2) had $\frac{S}{k^+}$; $\int_{z, w, w'}$ should be $\int_z$ (w, w' are gone); x' missing from $\int_{x, y, y'}$; $U^{ce}(x) / U^{c'e'}(x')$ have their colour indices in the opposite order to their partners

2. The w, w' integration — where the stray factor comes from

With $\beta = \frac{p^+}{S}$, the delta is $\delta^{(2)}[y - (1 - \beta)w - \beta z]$, so

$$w = \frac{y - \beta z}{1 - \beta}, \quad \text{Jacobian} = \frac{1}{(1 - \beta)^2} = \frac{S^2}{k^{+2}} \text{ per delta,}$$

$$w - z = \frac{y - z}{1 - \beta} \implies \frac{(w - z)^m}{(w - z)^2} = (1 - \beta) \frac{(y - z)^m}{(y - z)^2} \implies \frac{k^+}{S} \text{ per kernel,}$$

$$w' - w = \frac{y' - y}{1 - \beta} = \frac{S}{k^+} (y' - y) \implies e^{-ik \cdot (w' - w)} = e^{-i\frac{S}{k^+} k \cdot (y' - y)}.$$

Net from two deltas and two kernels: $\frac{S^4}{k^{+4}} \cdot \frac{k^{+2}}{S^2} = \frac{S^2}{k^{+2}}$. Together with the $\frac{p^+ k^+}{S^4}$ already in (0.1) this gives $\frac{p^+}{k^+ S^2}$ in front of the bare three-term tensor. Comparing that with your $\frac{1}{k^+} \cdot \frac{S}{k^+} \times [\text{bracket}]$, every one of the three structures comes out with the *same* ratio $\frac{S}{k^+}$ — so the bracket is right and the extra $\frac{(k^+ + p^+)}{k^+}$ must go.

Corrected (0.2)/(0.3):

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{N_c}{k^+} \int_{\Lambda}^{\vee-k^+} \frac{dp^+}{2\pi} \int_z \int_{x,x',y,y'} e^{-i \frac{s}{k^+} k \cdot (y' - y)} \frac{(y - z)^m (y' - z)^{m'}}{(y - z)^2 (y' - z)^2} X^k X'^{k'} \mathcal{B}^{mm'kk'} \times \mathcal{U}$$

$$\mathcal{B}^{mm'kk'} = \frac{d_{\perp} p^+}{S^2} \delta^{km} \delta^{k'm'} + \frac{2}{k^+} (\delta^{mk'} \delta^{m'k} - \delta^{m'k'} \delta^{km}) + \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right) \delta^{mm'} \delta^{kk'}$$

with $\mathcal{U} = [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)$.

3. Where $d_{\perp}$ goes — in detail

(a) Inside $\mathcal{G}^{mm'}$ it cancels exactly

Your (0.12) carries $\frac{\delta^{mm'}}{d_{\perp}}$; master (0.13) carries $\frac{d}{2}$:

$$\frac{1}{d_{\perp}} \cdot \frac{d}{2} \Gamma \left[1 - \frac{d}{2} \right] = \frac{1}{2} \Gamma[\epsilon] = \frac{\Gamma[1 + \epsilon]}{2\epsilon}$$

which is the $(1 - \epsilon)$ sitting next to $\Gamma[\epsilon]$ in your (0.16). Equivalently, in (0.18),

$$\frac{1}{d_{\perp}} \left(\frac{1}{\epsilon} - 1 \right) = \frac{1}{2(1 - \epsilon)} \cdot \frac{1 - \epsilon}{\epsilon} = \frac{1}{2\epsilon} \quad \text{exactly.}$$

No $O(\epsilon)$ remainder, so $d_{\perp}$ leaves no trace in $\mathcal{G}$ itself. That is why (0.20) has a clean $\frac{\pi \delta^{mm'}}{2}$.

(b) At the contraction it returns, twice

Writing $\mathcal{G}^{mm'} = \frac{\pi \delta^{mm'}}{2} \mathcal{L} - \pi \frac{r^m r^{m'}}{r^2}$ and using $\epsilon \mathcal{L} \rightarrow 1$, $\epsilon \times (\text{finite}) \rightarrow 0$:

$$\delta^{mm'} \mathcal{G}^{mm'} = \frac{\pi d_{\perp}}{2} \mathcal{L} - \pi = \pi(1 - \epsilon) \mathcal{L} - \pi = \pi(\mathcal{L} - 2),$$

$$d_{\perp} X_m X'_m \mathcal{G}^{mm'} = 2(1 - \epsilon) \left[\frac{\pi}{2} \mathcal{L} (X \cdot X') - \pi \frac{(X \cdot r)(X' \cdot r)}{r^2} \right] = \pi(\mathcal{L} - 1)(X \cdot X') - 2\pi \frac{(X \cdot r)(X' \cdot r)}{r^2}.$$

So $d_{\perp}$ is responsible for $\mathcal{L} \rightarrow \mathcal{L} - 1$ in the first bracket term and for one of the two units in $\mathcal{L} \rightarrow \mathcal{L} - 2$ in the third (the other unit is the trace of $-\pi \hat{r}^m \hat{r}^{m'}$). Setting $d_{\perp} = 2$ before contracting would give $\mathcal{L}$ and $\mathcal{L} - 1$ instead — wrong by $\pi(X \cdot X')$ in each.

4. The μ power and $\overline{\text{MS}}$: (0.19) and (0.20) disagree

Expanding the prefactor of (0.19) as written, with $\mu^{+2\epsilon}$,

$$\pi^{1-\epsilon} (r^2)^{-\epsilon} \mu^{2\epsilon} \mathcal{N}(\epsilon) = \pi \left[1 + \epsilon \left(2 - \gamma_E + \log \frac{\mu^2}{\pi r^2} \right) \right], \quad \mathcal{N}(\epsilon) = \frac{\Gamma[1 + \epsilon] \Gamma[1 - \epsilon]^2}{\Gamma[2 - 2\epsilon]},$$

$$\implies \mathcal{G}^{mm'} = \frac{\pi \delta^{mm'}}{2} \left[\frac{1}{\epsilon} + 2 - \gamma_E + \log \frac{\mu^2}{\pi r^2} \right] - \frac{\pi r^m r^{m'}}{r^2}.$$

Now $\mu_{\overline{\text{MS}}}^2 = 4\pi e^{-\gamma_E} \mu^2$, i.e. $\mu^2 = \frac{\mu_{\overline{\text{MS}}}^2 e^{\gamma_E}}{4\pi}$, gives

$$\log \frac{\mu^2}{\pi r^2} = \log \frac{\mu_{\overline{\text{MS}}}^2}{4\pi^2 r^2} + \gamma_E \quad \implies \quad \mathcal{G}^{mm'} = \frac{\pi \delta^{mm'}}{2} \left[\frac{1}{\epsilon} + 2 + \log \frac{\mu_{\overline{\text{MS}}}^2}{4\pi^2 r^2} \right] - \frac{\pi r^m r^{m'}}{r^2},$$

with the log **the other way up** from your (0.20). Which one you want is fixed by the power of μ :

you attach	then $\overline{\text{MS}}$ gives	log argument
$\mu^{+2\epsilon}$, as in (0.19)	$\frac{1}{\epsilon} + 2 + \log \frac{\mu_{\overline{\text{MS}}}^2}{4\pi^2 r^2}$	mass dimension 4 — only legitimate if the remaining $\int_{x,x',y,y'}$ are also kept in d dimensions and supply the rest
$\mu^{-2\epsilon}$	$\frac{1}{\epsilon} + 2 - \log (4\pi^2 \mu_{\overline{\text{MS}}}^2 r^2)$, as in your (0.20)	dimensionless $\checkmark$ — but then the scale must be redefined the other way, $\mu^2 \rightarrow 4\pi e^{-\gamma_E} \mu^2$

Since the other four transverse integrals in this row are finite and can be taken at $d_\perp = 2$, the self-contained choice is $\mu^{-2\epsilon}$ — i.e. **your (0.20) is the line to keep, and (0.19) should carry $\mu^{-2\epsilon}$** . Everything below uses

$$\mathcal{L} \equiv \frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu_{\overline{\text{MS}}}^2 r^2), \quad \mathcal{G}^{mm'} = \frac{\pi \delta^{mm'}}{2} \mathcal{L} - \frac{\pi r^m r^{m'}}{r^2}.$$

(If you keep $\mu^{+2\epsilon}$ instead, every $\mathcal{L}$ below simply becomes $\frac{1}{\epsilon} + 2 + \log \frac{\mu_{\overline{\text{MS}}}^2}{4\pi^2 r^2}$; nothing else changes.)

5. The contraction — finishing (0.21)

$\mathcal{G}^{mm'}$ is **symmetric**, so the two index pairings in the $\frac{2}{k^+}$ term give the same thing:

$$\delta^{mk'} \delta^{m'k} X^k X'^{k'} \mathcal{G}^{mm'} = X'^m X^{m'} \mathcal{G}^{mm'} = X^m X'^{m'} \mathcal{G}^{mm'} = \delta^{m'k'} \delta^{km} X^k X'^{k'} \mathcal{G}^{mm'}$$

So the entire $\frac{2}{k^+} (\delta^{mk'} \delta^{m'k} - \delta^{m'k'} \delta^{km})$ term drops out identically. It never contributes to this row.

What is left is two structures. Writing $\mathcal{G}_{XX'} \equiv X^m X'^{m'} \mathcal{G}^{mm'}$:

$$\frac{1}{k^+} \mathcal{B}^{mm'kk'} \mathcal{G}^{mm'} X^k X'^{k'} = \underbrace{\frac{p^+}{k^+ S^2} d_\perp \mathcal{G}_{XX'}}_{\text{first term}} + \underbrace{\left(\frac{p^+}{k^{+3}} + \frac{1}{p^+ k^+} \right) \delta^{mm'} \mathcal{G}^{mm'} (X \cdot X')}_{\text{third term}}$$

$$\mathcal{C} = \frac{\pi p^+}{k^+ S^2} \left[(\mathcal{L} - 1)(X \cdot X') - \frac{2(X \cdot r)(X' \cdot r)}{r^2} \right] + \pi \left(\frac{p^+}{k^{+3}} + \frac{1}{p^+ k^+} \right) (\mathcal{L} - 2)(X \cdot X')$$

Note $\frac{p^+}{k^{+3}} + \frac{1}{p^+ k^+} = \frac{p^{+2} + k^{+2}}{p^+ k^{+3}}$, which is exactly $\frac{1}{k^+} \left[\frac{S^2}{p^{+k^+2}} - \frac{2}{k^+} \right]$ — i.e. the $-\frac{2}{k^+}$ of the vanishing middle term has already been absorbed, as it must be.

6. The p^+ integration

Change variable to $\sigma = \frac{S}{k^+} = 1 + \frac{p^+}{k^+}$, so $dp^+ = k^+ d\sigma$ and $\sigma \in [1 + \lambda, \sigma_V]$ with $\lambda = \frac{\Lambda}{k^+}$, $\sigma_V = \frac{V}{k^+}$:

$$\frac{p^+}{k^+ S^2} = \frac{1}{k^{+2}} \frac{\sigma - 1}{\sigma^2}, \quad \frac{p^+}{k^{+3}} + \frac{1}{p^+ k^+} = \frac{1}{k^{+2}} \left[(\sigma - 1) + \frac{1}{\sigma - 1} \right].$$

The phase is $e^{-i\sigma\kappa}$ with $\kappa = k_\perp \cdot r$ and $\mathcal{L}$ does not depend on σ , so with $\frac{k^+}{2\pi} \cdot \pi \cdot \frac{1}{k^{+2}} = \frac{1}{2k^+}$ the whole p^+ integral reduces to two masters:

$$\int \frac{dp^+}{2\pi} e^{-i\sigma\kappa} \mathcal{C} = \frac{1}{2k^+} \left[A J_1(\kappa) + B J_2(\kappa) \right],$$

$$A = (\mathcal{L} - 1)(X \cdot X') - \frac{2(X \cdot r)(X' \cdot r)}{r^2}, \quad B = (\mathcal{L} - 2)(X \cdot X').$$

J_1 — **convergent, closed form**

$$J_1 = \int_1^\infty d\sigma \frac{\sigma - 1}{\sigma^2} e^{-i\sigma\kappa} = \int_1^\infty \left(\frac{1}{\sigma} - \frac{1}{\sigma^2} \right) e^{-i\sigma\kappa} d\sigma = E_1(i\kappa) - E_2(i\kappa)$$

and with $E_2(z) = e^{-z} - z E_1(z)$,

$$J_1(\kappa) = (1 + i\kappa) E_1(i\kappa) - e^{-i\kappa}$$

(verified against oscillatory quadrature to 8 digits at $\kappa = 0.3, 1.0, 2.5$). Small κ : $J_1 \rightarrow -\gamma_E - \log(i\kappa) - 1$, so this term carries a $\log \kappa$ but no Λ .

J_2 — **this is where the rapidity log lives**

Substitute $t = \sigma - 1$ and use $(\sigma - 1) + \frac{1}{\sigma - 1} = t + \frac{1}{t}$:

$$J_2 = e^{-i\kappa} \left[\underbrace{\int_{\lambda}^T t e^{-i\kappa t} dt}_{\text{large-}p^+} + \underbrace{\int_{\lambda}^T \frac{e^{-i\kappa t}}{t} dt}_{\text{small-}p^+} \right], \quad T = \sigma_V - 1,$$

$$= e^{-i\kappa} \left[\frac{e^{-i\kappa T}(1 + i\kappa T) - e^{-i\kappa \lambda}(1 + i\kappa \lambda)}{\kappa^2} + E_1(i\kappa \lambda) - E_1(i\kappa T) \right]$$

(this closed form reproduces direct quadrature to 8 digits). Now take $\lambda \rightarrow 0$ with $E_1(x) \rightarrow -\gamma_E - \log x$:

$$E_1(i\kappa \lambda) \rightarrow \log \frac{k^+}{\Lambda} - \gamma_E - \log(i\kappa) = \log \frac{k^+}{\Lambda} - \gamma_E - \log \kappa - i\frac{\pi}{2}.$$

The $\vee$ -dependent pieces ($e^{-i\kappa T}$, $E_1(i\kappa T)$) oscillate in κ with $T \rightarrow \infty$ and are killed by Riemann–Lebesgue once the remaining $\int_{x,x',y,y'}$ are done. What survives is

$$J_2(\kappa) = e^{-i\kappa} \left[\log \frac{k^+}{\Lambda} - \gamma_E - \log(i\kappa) - \frac{1}{\kappa^2} \right] + (\text{oscillating, drops})$$

Why the phase's p^+ dependence is not optional. If you keep $e^{-ik \cdot (y'-y)}$ as in (0.2)–(0.3), the $\frac{p^+}{k^+}$ half of the third term integrates to $\int^{\vee} dp^+ \frac{p^+}{k^+} = \frac{\vee^2}{2k^+}$ — a quadratic divergence in the longitudinal cutoff, which cannot be right. Restoring σ in the exponent turns it into the bounded oscillating term above. That is the cross-check that the $\frac{S}{k^+}$ belongs in the phase and not as a prefactor.

7. The finished result

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{N_c}{2k^+} \int_{x,x',y,y'} \left\{ J_1(\kappa) A(r) + J_2(\kappa) B(r) \right\} [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)$$

$$A = (\mathcal{L} - 1) (X \cdot X') - \frac{2(X \cdot r)(X' \cdot r)}{r^2}, \quad B = (\mathcal{L} - 2) (X \cdot X'), \quad \mathcal{L} = \frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu_{\overline{\text{MS}}}^2 r^2)$$

with $r = y' - y$, $\kappa = k_{\perp} \cdot r$, $X = \frac{x-y}{(x-y)^2}$, $X' = \frac{x'-y'}{(x'-y')^2}$.

Reading off the divergences

- $\frac{1}{\epsilon}$ with coefficient $(X \cdot X') [J_1(\kappa) + J_2(\kappa)]$ — the transverse UV of the z integral, to be removed by the $\overline{\text{MS}}$ counterterm / charge renormalisation.
- $\log \frac{k^+}{\Lambda}$ from J_2 , with coefficient $e^{-i\kappa} (\mathcal{L} - 2) (X \cdot X')$ — the rapidity divergence, absorbed into the JIMWLK evolution of $[U(y') - U(x')] [U(y) - U(x)]$.
- $\frac{1}{\epsilon} \log \frac{k^+}{\Lambda}$ — the same mixed UV $\times$ rapidity term this row produces as the running-coupling row did. It cannot stand alone; check it against the companion rows of Group II in the same scheme.
- **No collinear pole.** The would-be $r \rightarrow 0$ singularity is only $\log r^2$, which is what $\mathcal{L}$ carries; J_1, J_2 contribute $\log \kappa = \log(k_{\perp} \cdot r)$, so the μ dependence combines into $\log(\mu_{\overline{\text{MS}}}^2 / k_{\perp}^2)$ once the r integral is done — no extra $\frac{1}{\epsilon}$.

8. Checklist

- (0.2): delete the overall $\frac{(k^+ + p^+)}{k^+}$; the prefactor is $\frac{1}{k^+}$.
- (0.2), (0.3), (0.21): the phase is $e^{-i \frac{(k^+ + p^+)}{k^+} k \cdot (y' - y)}$, written once.
- (0.3): $\int_{z,w,w'} \rightarrow \int_z$; add x' to $\int_{x,y,y'}$; remove the stray $\frac{p^+}{k^+}$; fix the colour index order in $U^{ee}(x)$, $U^{e'e'}(x')$.
- (0.19): $\mu^{2\epsilon} \rightarrow \mu^{-2\epsilon}$ (then (0.20) is right as written).
- (0.21): keep $d_{\perp} = 2 - 2\epsilon$ through the contraction — it shifts $\mathcal{L} \rightarrow \mathcal{L} - 1$ and $\mathcal{L} \rightarrow \mathcal{L} - 2$.
- The $\frac{2}{k^+}$ term contributes zero; drop it.

All of the above verified in `src/rowG2_check.py` (tensor product in $D = 3, 4, 5$; w, w' Jacobians; the $d_{\perp}$ identity; J_1 against oscillatory quadrature; J_2 against direct quadrature).